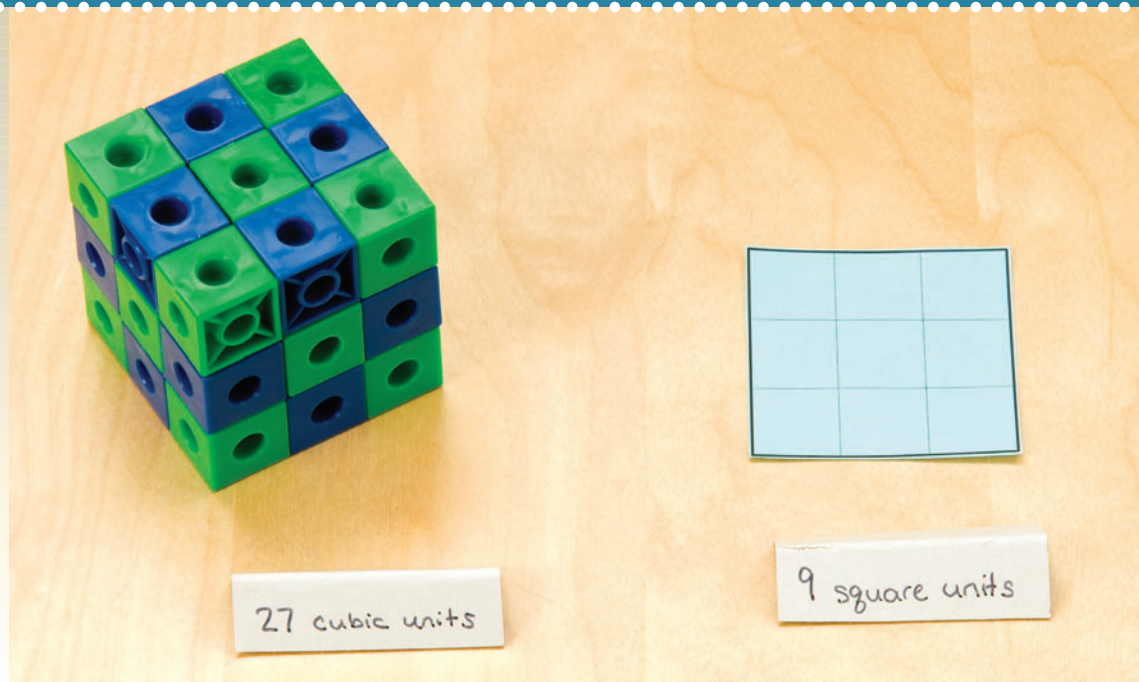


4.1

Estimating Roots

LESSON FOCUS

Explore decimal representations of different roots of numbers.



Make Connections

Since $3^2 = 9$, 3 is a square root of 9.

We write: $3 = \sqrt{9}$

Since $3^3 = 27$, 3 is the cube root of 27.

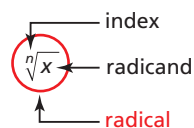
We write: $3 = \sqrt[3]{27}$

Since $3^4 = 81$, 3 is a fourth root of 81.

We write: $3 = \sqrt[4]{81}$

How would you write 5 as a square root?

A cube root? A fourth root?



Construct Understanding

TRY THIS

Work with a partner.

You will need a calculator to check your estimates.

- A.** Write the two consecutive perfect squares closest to 20. Estimate the value of $\sqrt{20}$. Square your estimate. Use this value to revise your estimate. Keep revising your estimate until the square of the estimate is within 1 decimal place of 20.
- B.** Write the two consecutive perfect cubes closest to 20. Estimate the value of $\sqrt[3]{20}$. Cube your estimate. Use this value to revise your estimate. Keep revising your estimate until the cube of the estimate is within 1 decimal place of 20.
- C.** Write the two consecutive perfect fourth powers closest to 20. Use a strategy similar to that in Steps A and B to estimate a value for $\sqrt[4]{20}$.
- D.** Copy and complete this table. Use the strategies from Steps A to C to determine the value of each radical.

Radical	Value	Is the Value Exact or Approximate?
$\sqrt{16}$	4	Exact
$\sqrt{27}$	5.1962	Approximate
$\sqrt{\frac{16}{81}}$	$\frac{4}{9}$ or 0.4	Exact
$\sqrt{0.64}$		
$\sqrt[3]{16}$		
$\sqrt[3]{27}$		
$\sqrt[3]{\frac{16}{81}}$		
$\sqrt[3]{0.64}$		
$\sqrt[3]{-0.64}$		
$\sqrt[4]{16}$		
$\sqrt[4]{27}$		
$\sqrt[4]{\frac{16}{81}}$		
$\sqrt[4]{0.64}$		

Choose 3 different radicals.

Extend then complete the table for these radicals.

- E.** How can you tell if the value of a radical is a rational number? What strategies can you use to determine the value of the radical?
- F.** How can you tell if the value of a radical is *not* a rational number? What strategies can you use to estimate the value of the radical?

Assess Your Understanding

- Give 4 examples of radicals. Use a different index for each radical.
 - Identify the radicand and index for each radical.
 - Explain the meaning of the index of each radical.
- Evaluate each radical. Justify your answer.
 - $\sqrt{36}$
 - $\sqrt[3]{8}$
 - $\sqrt[4]{10\,000}$
 - $\sqrt[5]{-32}$
 - $\sqrt[3]{\frac{27}{125}}$
 - $\sqrt{2.25}$
 - $\sqrt[3]{0.125}$
 - $\sqrt[4]{625}$
- Estimate the value of each radical to 1 decimal place. What strategy did you use?
 - $\sqrt{8}$
 - $\sqrt[3]{9}$
 - $\sqrt[4]{10}$
 - $\sqrt{13}$
 - $\sqrt[3]{15}$
 - $\sqrt[4]{17}$
 - $\sqrt{19}$
 - $\sqrt[3]{20}$
- What happens when you attempt to determine the square root of a number such as -4 ? Explain the result.
 - For which other radical indices do you get the same result with a negative radicand, as in part a?
 - When a radicand is negative:
 - Which types of radicals can be evaluated or estimated?
 - Which types of radicals cannot be evaluated or estimated?
- For each number below, write an equivalent form as:
 - a square root
 - a cube root
 - a fourth root
 - 2
 - 3
 - 4
 - 10
 - 0.9
 - 0.2
- Choose values of n and x so that $\sqrt[n]{x}$ is:
 - a whole number
 - a negative integer
 - a rational number
 - an approximate decimal
 Verify your answers.

